1. Compute the total number of parameters for additive attention.

Compute the total number of parameters for additive attention for n queries and m key-value pairs, where queries are of length q, keys are of length k, and values are of length v. Give the answer symbolically. Numerical application: n=2, m=3, q=k=2, v=3.

Compute the total number of parameters for additive attention for n queries and m key-value pairs, where queries are of length q, keys are of length k, and values are of length v. Give the answer symbolically. Numerical application:
$n = 2, m = 3, q = k = 2, v = 3.$

Soluție:

1. n queries -> length q
m key-value pairs
↳ length k ↳ length v

$n=2$
$m=3$
$q=k=2$
$v=3$

Teorie:

- Additive attention: $a(q,k) = w_v^T \tanh(W_q q + W_k k)$
- $f = f(q,(k_1,v_1),\dots,(k_m,v_m)) = \sum_{i=1}^{m} \alpha(q,k_i) v_i$, $v_i \in \mathbb{R}^v$

$\alpha(q,k_i) = \dfrac{\exp(a(q,k_i))}{\sum_{j=1}^{m} \exp(a(q,k_j))} \in \mathbb{R}$

$W_q \in \mathbb{R}^{h\times q}$ => $h\times q$ params
$W_k \in \mathbb{R}^{h\times k}$ => $h\times k$ params
$w_i \in \mathbb{R}^{h}$ => h params

=> $a = hq + hk + h = h(q+k+1)$ params

$\alpha = \dfrac{\exp a}{\sum_{1}^{m} \exp(a)}$ => $a + m\cdot a = h(q+k+1)(1+m)$

$f = m\cdot h\cdot(q+k+1)(1+m)$
$n = 2$ queries => $2\cdot f$

Ca date din pb:
$n=2$, $m=3$, $q=k=2$, $v=3$ => $2\cdot 3\cdot h(2+2+1)(1+3) = 6\cdot h\cdot 5\cdot 4 = 120h$, h = nr of hidden layers

2. Compute symbolically the total number of parameters of a GRU layer

Compute symbolically the total number of parameters of a GRU layer, knowing that the number of examples is n, the number of inputs is d, and the number of hidden units is h. Numerical application: n=2, d=3, h=4

Compute symbolically the total number of parameters of a GRU layer, knowing that the number of examples is n, the number of inputs is d, and the number of hidden units is h. Numerical application: $n = 2, d = 3, h = 4$.

Soluție:

2. n - nr. examples
d - inputs
h - hidden layers

$n=2$
$d=3$
$h=4$

pdom. GRU layer

Teorie:

$$R_t = \sigma(X_t W_{xr} + H_{t-1} W_{hr} + b_r)$$
$$Z_t = \sigma(X_t W_{xz} + H_{t-1} W_{hz} + b_z)$$
$$\tilde{H}_t = \tanh(X_t W_{xh} + (R_t \odot H_{t-1}) W_{hh} + b_h)$$
$$H_t = Z_t \odot H_{t-1} + (1 - Z_t) \odot \tilde{H}_t$$

Unde: $W_{xr}, W_{xz}, W_{xh} \in \mathbb{R}^{d \times h}$ => $d \times h$ pdom
$W_{hr}, W_{hz}, W_{hh} \in \mathbb{R}^{h \times h}$ => $h \times h$ pdom
$b_r, b_z, b_h \in \mathbb{R}^{1 \times h}$ => $1 \times h$ pdoms

Nr. pdom: $g \cdot (dh + hh + h)$
$d = 3$
$h = 4$
$n = 2$

3?

$$\Rightarrow n \cdot g(dh + hh + h) =$$
$$= 2 \cdot 3 (3 \cdot 4 + 4 \cdot 4 + 4)$$
$$= 6 \cdot (12 + 16 + 4)$$
$$= 192 \text{ pdom.}$$

3. Support vectors for a hard-margin SVM

Assume we have a training set for two-class classification in two dimensions that contains seven sample points: points (1,1), (2,2), (3,1) with label +1 and points (4,5), (5,7), (6,5), (7,7) with label -1. Which points are the support vectors for a hard-margin SVM?

Assume we have a training set for two-class classification in two dimensions that contains seven sample points: points $(1, 1), (2, 2), (3, 1)$ with label $+1$ and points $(4, 5), (5, 7), (6, 5), (7, 7)$ with label -1. Which points are the support vectors for a hard-margin SVM?

Soluție:

3. 2 class classification with 2 labels:

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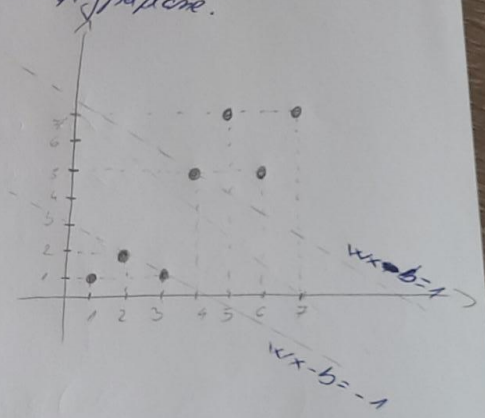
$(1,1), (2,2), (3,1) \to +1$
$(4,5), (5,7), (6,5), (7,7) \to -1$ } hard-margin SVM

- Hard margin equation: $\min\limits_{w, w_0} \frac{1}{2}\|w\|_2^2$ subject to $y^{(i)}(w^T x^{(i)} + w_0) \geq 1$, $i = \overline{1,m}$

- Support vectors are the data points nearest to the hyperplane.

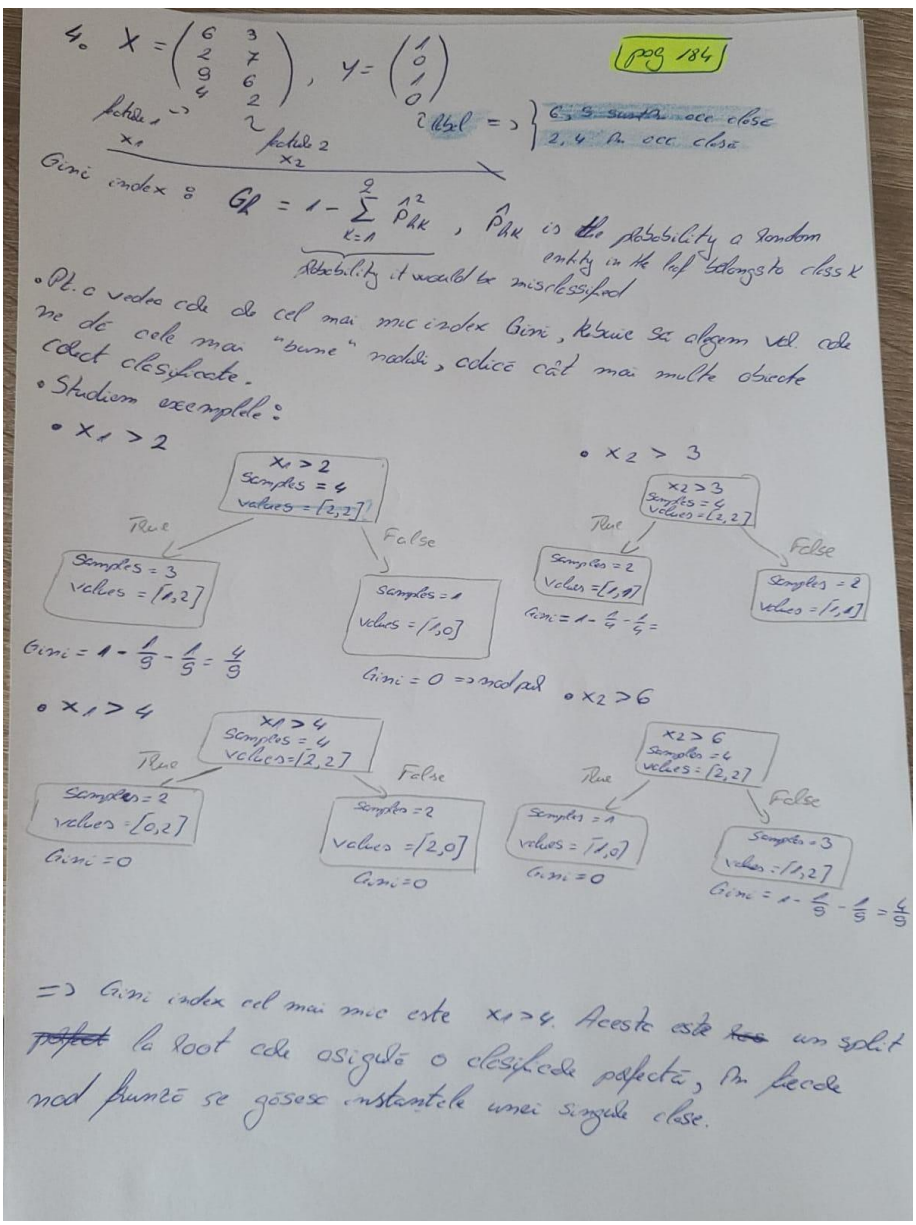
- Am reprezentat grafic punctele noastre => punctele care sunt support vectors sunt: $(2,2)$, $(3,1)$ } label $+1$; $(4,5)$ } label -1



4. Decision trees. Which of the following splits at the root node gives the smallest entropy:

Consider training a classification decision tree given a design matrix $X = \begin{bmatrix} 6 & 3 \\ 2 & 7 \\ 9 & 6 \\ 4 & 2 \end{bmatrix}$ and labels $y = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$. Let x_1 denote feature 1, corresponding to the first column of X, and let x_2 denote feature 2, corresponding to the second column of X. Which of the following splits at the root node gives the smallest entropy: $x_1 > 2$, $x_1 > 4$, $x_2 > 3$, $x_2 > 6$? Motivate your answer.

Solutie:



4. $X = \begin{pmatrix} 6 & 3 \\ 2 & 7 \\ 9 & 6 \\ 4 & 2 \end{pmatrix}$, $y = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$

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feature 1 → x_1 ; feature 2 → x_2

label ⇒ { 6, 9 sunt în acc clase ; 2, 4 în acc clasă

Gini index: $G_k = 1 - \sum_{k=1}^{2} \hat{p}_{mk}^2$, $\hat{p}_{mk}$ is the probability a random entity in the leaf belongs to class k

probability it would be misclassified

• Pt. a vedea care de cel mai mic index Gini, trebuie să alegem vel. care ne dă cele mai "bune" noduri, adică cât mai multe obiecte corect clasificate.

• Studiem exemplele:

• $x_1 > 2$

$x_1 > 2$, Samples = 4, values = [2,2]

True: Samples = 3, values = [1,2]

$Gini = 1 - \frac{1}{9} - \frac{1}{9} = \frac{4}{9}$

False: Samples = 1, values = [1,0]

Gini = 0 ⇒ nod pur

• $x_2 > 3$

$x_2 > 3$, Samples = 4, values = [2,2]

True: Samples = 2, values = [1,1]

$Gini = 1 - \frac{1}{4} - \frac{1}{4} =$

False: Samples = 2, values = [1,1]

• $x_1 > 4$

$x_1 > 4$, Samples = 4, values = [2,2]

True: Samples = 2, values = [0,2]

Gini = 0

False: Samples = 2, values = [2,0]

Gini = 0

• $x_2 > 6$

$x_2 > 6$, Samples = 4, values = [2,2]

True: Samples = 1, values = [1,0]

Gini = 0

False: Samples = 3, values = [1,2]

$Gini = 1 - \frac{4}{9} - \frac{1}{9} = \frac{4}{9}$

⇒ Gini index cel mai mic este $x_1 > 4$. Acesta este ~~ceea~~ un split ~~perfect~~ la root care asigură o clasificare perfectă, în fiecare nod frunză se găsesc instanțele unei singure clase.

5. Use PCA to reduce the dimension from 2 to 1, to compute the first principal component.

Assume we have the design matrix

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{bmatrix}.$$

Use PCA to reduce the dimension from 2 to 1, i.e., to compute the first principal component.

Solutie:

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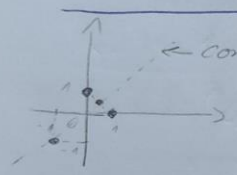
5. $X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{pmatrix}$

- PCA to reduce dim. from 2 to 1
- compute the first principal component



- simple set mean: $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x^{(i)}$

$$\bar{x} = \frac{1}{3}\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ -1 \end{pmatrix}\right)$$

$$\bar{x} = \frac{1}{3} \cdot \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

- covariance matrix: $\Sigma = \frac{1}{n} \sum_{i=1}^{n} (x^{(i)} - \bar{x})(x^{(i)} - \bar{x})^T$

$$\Sigma = \frac{1}{3}\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}(1 \ 0) + \begin{pmatrix} 0 \\ 1 \end{pmatrix}(0 \ 1) + \begin{pmatrix} -1 \\ -1 \end{pmatrix}(-1 \ -1)\right)$$

$$\Sigma = \frac{1}{3}\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\right) = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

- direction w_1 = $(x \ x)$

$$\det(\Sigma - \lambda I_2) = 0 \Rightarrow \begin{vmatrix} \frac{2}{3} - \lambda & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} - \lambda \end{vmatrix} = \left(\frac{2}{3} - \lambda\right)^2 - \frac{1}{9} = 0$$

$$\Rightarrow \left(\frac{2}{3} - \lambda\right)^2 = \frac{1}{9} \Rightarrow \frac{2}{3} - \lambda = \pm \frac{1}{3} \Rightarrow \lambda = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$$

$$\Rightarrow \lambda = \frac{2}{3} + \frac{1}{3} = \frac{3}{3} = 1$$

Se ia cea maxima

$\Sigma w_1 = \lambda w_1 \Rightarrow$ Notam $w_1 = \begin{pmatrix} a \\ b \end{pmatrix}$

$$\begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = 1\begin{pmatrix} a \\ b \end{pmatrix} \Rightarrow \begin{cases} \frac{2}{3}a + \frac{1}{3}b = a \\ \frac{1}{3}a + \frac{2}{3}b = b \end{cases} \Rightarrow a = b \Rightarrow w_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

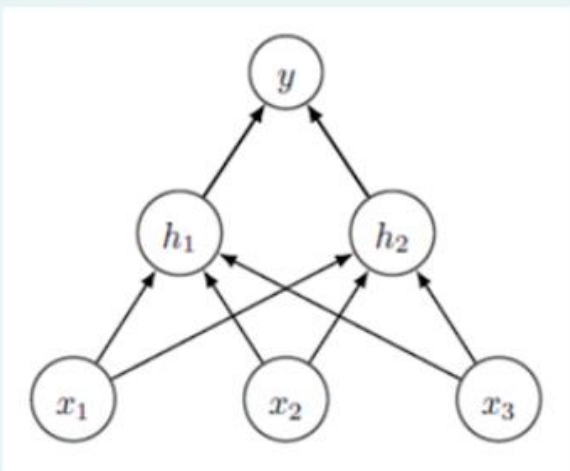
$\frac{1}{3}a = \frac{1}{3}b$

plina comp. principale

6. The following graph shows the structure … compute symbolically.

The following graph shows the structure of a simple neural network with a single hidden layer. The input layer consists of three dimensions $x = (x_1, x_2, x_3)$. The hidden layer includes two units $h = (h_1, h_2)$. The output layer includes one unit y. We ignore bias terms for simplicity. We use sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$ as activation function for the hidden layer and the identity for the output layer. Moreover, denote by $l(y,t) = \frac{1}{2}(y-t)^2$ the loss function. Here, t is the target value for the output unit y. Denote by W and V weight matrices connecting input and hidden layer, and hidden layer and output, respectively. Compute symbolically $\frac{\partial l}{\partial W}$. Numerical application:

$$W = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}, V = \begin{bmatrix} 0 & 1 \end{bmatrix}, x = (1, 2, 1), t = 1.$$



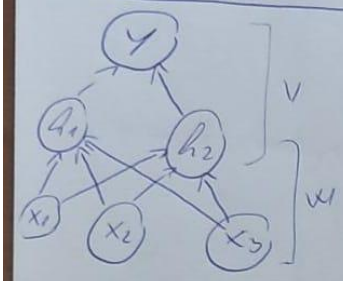
Solutie:

6. activation function: $\sigma(z) = \frac{1}{1+e^{-z}}$ (2) $\boxed{\text{pag 662}}$

Loss function: $l(y,t) = \frac{1}{2}(y-t)^2$, t = target val. for the output unit y

W, V - weight matrices

$W = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$, $V = [0 \quad 1]$, $x = (1, 2, 1)$, $t = 1$



$H = W \cdot x$ (1)

$y = \sigma(W \cdot x) \cdot V$, we ignore the biases (3)

$\Rightarrow l(y,t) = \frac{1}{2}(V \cdot \sigma(W \cdot x) - t)^2$

$\Rightarrow \frac{\partial l}{\partial W} = \frac{1}{2} \cdot 2 (V \cdot \sigma(Wx) - t) \cdot \sigma(Wx)$

$= (V \cdot \sigma(Wx) - t) \cdot \sigma(Wx)$

(1) $\Rightarrow H = W \cdot x = \begin{pmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$

(2) $\Rightarrow \sigma(H) = \sigma\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{1}{1+e^{-2}} \\ \frac{1}{1+e} \end{pmatrix}$ $\xRightarrow{(3)}$

$V^T = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$\Rightarrow y = (0 \quad 1) \begin{pmatrix} \frac{1}{1+e^{-2}} \\ \frac{1}{1+e} \end{pmatrix} = \frac{1}{1+e} = [0{,}265]$

$\frac{\partial l}{\partial W} = V^T ([0{,}265] - 1) \odot \sigma'(z)) \cdot x^T$

$= \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \cdot [-0{,}731] \odot \sigma'(z) \right) \cdot x^T$

$\sigma'(z) = \frac{e^{-z}}{(1+e^{-z})^2} = \begin{bmatrix} 0{,}105 \\ 0{,}197 \end{bmatrix}$

$\frac{\partial l}{\partial W} = \left(\begin{bmatrix} 0 \\ -0{,}731 \end{bmatrix} \odot \begin{bmatrix} 0{,}105 \\ 0{,}197 \end{bmatrix} \right) \cdot [1 \quad 2 \quad 1] = \begin{bmatrix} 0 \\ -0{,}144 \end{bmatrix} \cdot [1 \quad 2 \quad 1]$

$= \begin{bmatrix} 0 & 0 & 0 \\ -0{,}144 & -0{,}288 & -0{,}144 \end{bmatrix}$

7. What is the distance between clusters using single linkage, complete linkage?

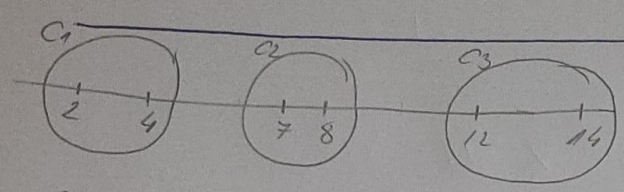
Consider a dataset containing six one-dimensional points: $\{2, 4, 7, 8, 12, 14\}$. After three iterations of hierarchical clustering using Euclidean distance between points, we get the 3 clusters: $C_1 = \{2, 4\}$, $C_2 = \{7, 8\}$, and $C_3 = \{12, 14\}$. What is the distance between clusters C_1 and C_2 using single linkage? What is the distance between clusters C_1 and C_2 using complete linkage?

Solutie:

7. 6 1-D $\{2, 4, 7, 8, 12, 14\}$ pag 386

3 clusters: $C_1 = \{2, 4\}$; $C_2 = \{7, 8\}$, $C_3 = \{12, 14\}$

Dist. between C_1 and C_2 with → single linkage
→ complete linkage



- Single linkage: $d_{SL}(C_1, C_2) = \min_{\substack{i \in C_1 \\ i' \in C_2}} d_{i,i'}$

$\Rightarrow \underset{\substack{i \in C_1 \\ i' \in C_2}}{d_{i,i'}} = \{|2-7|, |2-8|, |4-7|, |4-8|\} = \{5, 6, 3, 4\} \Rightarrow$

$\Rightarrow d_{SL}(C_1, C_2) = 3$

- Complete linkage: $d_{CL}(C_1, C_2) = \max_{\substack{i \in C_1 \\ i' \in C_2}} d_{i,i'}$

$\Rightarrow d_{CL}(C_1, C_2) = 6$

8. Estimate the following linear regression model:

Assume we have a dataset $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(n)}, y^{(n)})\}$ of points in the 2D plane, and we want to estimate the following linear regression model:

$$\hat{y} = w_1 x + w_0.$$

Compute w_0, w_1 in terms of $x^{(1)}, y^{(1)}, x^{(2)}, y^{(2)}, \dots, x^{(n)}, y^{(n)}$. Numerical application: $(1, 1), (2, -1), (3, 2)$.

Solutie:

8. dataset: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(n)}, y^{(n)})\}$ pag 17 curs

$\hat{y} = w_1 x + w_0$

Fie $w = \begin{pmatrix} w_0 \\ w_1 \end{pmatrix}$, $x' = \begin{pmatrix} 1 \\ x \end{pmatrix}$ $\Rightarrow$ $\hat{y} = w^T \cdot x'$

Fie $X = \begin{pmatrix} 1 & x^{(1)} \\ \vdots & \vdots \\ 1 & x^{(n)} \end{pmatrix}$ matr. design

$\Rightarrow \hat{y} = X \cdot w$

Cmls

$\Rightarrow w = (X^T \cdot X)^{-1} X^T y \Rightarrow w = \begin{pmatrix} 1 & \cdots & 1 \\ x^{(1)} & \cdots & x^{(n)} \end{pmatrix} \begin{pmatrix} 1 & x^{(1)} \\ \vdots & \vdots \\ 1 & x^{(n)} \end{pmatrix}^{-1} \cdot \begin{pmatrix} 1 & \cdots & 1 \\ x^{(1)} & \cdots & x^{(n)} \end{pmatrix} \cdot y$

$\Rightarrow w = \begin{pmatrix} \underbrace{n}_{a} & \underbrace{\sum_{i=1}^{n} x^{(i)}}_{a} \\ \underbrace{\sum_{i=1}^{n} x^{(i)}}_{a} & \underbrace{\sum_{i=1}^{n} x^{(i)^2}}_{b} \end{pmatrix}^{-1} \cdot \begin{pmatrix} 1 & \cdots & 1 \\ x^{(1)} & \cdots & x^{(n)} \end{pmatrix} \cdot y$

$\Rightarrow w = \frac{1}{n \cdot b - a^2} \begin{pmatrix} b & -a \\ -a & n \end{pmatrix} \begin{pmatrix} 1 & \cdots & 1 \\ x^{(1)} & \cdots & x^{(n)} \end{pmatrix} \cdot y$

$w = \frac{1}{nb - a^2} \begin{pmatrix} b - a x^{(1)} & \cdots & b - a x^{(n)} \\ n \cdot x^{(1)} - a & \cdots & n \cdot x^{(n)} - a \end{pmatrix} \begin{pmatrix} y^{(1)} \\ \vdots \\ y^{(n)} \end{pmatrix}$

$\Rightarrow \begin{pmatrix} w_0 \\ w_1 \end{pmatrix} = \frac{1}{nb - a^2} \begin{pmatrix} \sum_{i=1}^{n} y^{(i)} (b - a x^{(i)}) \\ \sum_{i=1}^{n} y^{(i)} (n \cdot x^{(i)} - a) \end{pmatrix}$

$\Rightarrow w_0 = \frac{1}{nb - a^2} \cdot \sum_{i=1}^{n} y^{(i)} (b - a x^{(i)})$

$w_1 = \frac{1}{nb - a^2} \cdot \sum_{i=1}^{n} y^{(i)} (n \cdot x^{(i)} - a)$

$n = 3$

$x^{(1)} = 1$ $\quad y^{(1)} = 1$

$x^{(2)} = 2$ $\quad y^{(2)} = -1$

$x^{(3)} = 3$ $\quad y^{(3)} = 2$

$\Rightarrow a = \sum_{i=1}^{3} x^{(i)} = 1 + 2 + 3 = 6$

$b = \sum_{i=1}^{3} x^{(i)^2} = 1 + 4 + 9 = 14$

$\Rightarrow$

$\Rightarrow w_0 = \frac{1}{3 \cdot 14 - 36} \cdot \sum_{i=1}^{3} y^{(i)} (14 - 6 x^{(i)})$

$w_0 = \frac{1}{6} \cdot (1(14 - 6) + (-1)(14 - 12) + 2(14 - 18))$

$w_0 = \frac{1}{6} \cdot (8 - 2 - 8) = -\frac{2}{6} = -\frac{1}{3}$

$\Rightarrow w_1 = \frac{1}{6} \cdot (1(3 \cdot 1 - 6) - 1(3 \cdot 2 - 6) + 2(3 \cdot 3 - 6))$

$w_1 = \frac{1}{6} (-3 - 0 + 6) = \frac{3}{6} = \frac{1}{2}$

9. Assume we have an input image whose shape is 128 x 128 x 3.

Assume we have an input image whose shape is $128 \times 128 \times 3$. Compute the output size and number of parameters of each of the following possible settings for the first convolutional layer, given the input has the shape described above.

(a) The first convolutional layer has a stride of **1**, a filter size of **3**, input padding of **0**, and **64** filters.

(b) The first convolutional layer has a stride of **1**, a filter size of **5**, input padding of **2**, and **16** filters.

(c) The first convolutional layer has a stride of **2**, a filter size of **2**, input padding of **0**, and **32** filters.

Solutie:

9. INPUT image $128 \times 128 \times 3$ pag 527

~~Output size?~~ ~~Convolutional~~

a) ~~CONV = K = N~~ : $s = 1$, $K = 3$, $p = 0$, 64 filters

Tolie

$\hookrightarrow$ $output_h = (n_h - K_h + \overset{2\cdot}{p_h} + s_h)/s_h$! h e indice

$\hookrightarrow$ $output_w = (n_w - K_w + \overset{2\cdot}{p_w} + s_w)/s_w$! w e indice

$p_h = p_w = K_h - 1 = K_w - 1$

pg 563

$\Rightarrow$ $output_h = output_w = (128 - 3 + 0 + 1)/1 = 126$ $\Rightarrow$ output $126 \times 126 \times 3$

• parametri un filtru: $\underset{\text{bias}}{1} + \underset{\text{kernel}}{3 \times 3} = 10$ $\Rightarrow$ $10 \cdot 64 = 640$ parametri.

b) ~~POOL K~~ : $s = 1$, $K = 5$, $p = 2$, 16 filters

$\Rightarrow$ $output_h = output_w = 128 - 5 + 4 + 1 = 128$ $\Rightarrow$ $128 \times 128 \times 3$

• parametri un filtru: $1 + 5 \times 5 = 26$ $\Rightarrow$ $16 \cdot 26 = 416$ parametri

c) ~~FC N~~ : $s = 2$, $K = 2$, $p = 0$, 32 filters

$\Rightarrow$ $output_h = output_w = (128 - 2 + 0 + 2)/2 = 64$ $\Rightarrow$ output $64 \times 64 \times 3$

• parametri un filtru: $1 + 2 \times 2 = 5$ $\Rightarrow$ $32 \cdot 5 = 160$ parametri

10. K-means clustering algorithm

Assume we want to cluster 7 points into 3 clusters using k-means clustering algorithm. Suppose after first iteration clusters C1, C2, C3 contain the following 2-D points: C1: {(0,6),(6,0)}; C2:{(2,2),(4,4),(6,6)} and C3:{(5,5),(7,7)}

Assume we want to cluster 7 points into 3 clusters using the K-means clustering algorithm. Suppose after the first iteration, clusters C_1, C_2, and C_3 contain the following two-dimensional points:

(a) C_1 contains the 2 points: $\{(0,6),(6,0)\}$;

(b) C_2 contains the 3 points: $\{(2,2),(4,4),(6,6)\}$;

(c) C_3 contains the 2 points: $\{(5,5),(7,7)\}$

What are the cluster centers computed for these 3 clusters?

Solutie:

10. 7 points into 3 clusters

$C_1 : \{(0,6),(6,0)\}$

$C_2 : \{(2,2),(4,4),(6,6)\}$

$C_3 : \{(5,5),(7,7)$

K-Means clustering alg.

pag 348

Cluster centers ?

Cluster centers: $C_1 : \left(\frac{0+6}{2}, \frac{6+0}{2}\right) = (3,3)$

$C_2 = \left(\frac{2+4+6}{3}, \frac{2+4+6}{3}\right) = (4,4)$

$C_3 = \left(\frac{5+7}{2}, \frac{5+7}{2}\right) = (6,6)$

11. Compute symbolically the total number of parameters of an LSTM layer

Compute symbolically the total number of parameters of an LSTM layer, knowing that the number of examples is n, the number of inputs is d, and the number of hidden units is h. Numerical application: n=2, d=3, h=4.

Compute symbolically the total number of parameters of an LSTM layer, knowing that the number of examples is n, the number of inputs is d, and the number of hidden units is h. Numerical application: $n = 2, d = 3, h = 4$.

Solutie:

11. n - nr. of examples $n=2$ pag 698

d - inputs $d=3$ LSTM layer

h - hidden layers $h=4$

$$I_t = \sigma(X_t W_{xi} + H_{t-1} W_{hi} + b_i)$$
$$F_t = \sigma(X_t W_{xf} + H_{t-1} W_{hf} + b_f)$$
$$O_t = \sigma(X_t W_{xo} + H_{t-1} W_{ho} + b_o)$$
$$\tilde{C}_t = \tanh(X_t W_{xc} + H_{t-1} W_{hc} + b_c)$$

, where $W_{xi}, W_{xf}, W_{xo}, W_{xc} \in \mathbb{R}^{d \times h}$ => $d \times h$ param

$W_{hi}, W_{hf}, W_{ho}, W_{hc} \in \mathbb{R}^{h \times h}$ => $h \times h$ param

$b_i, b_f, b_o, b_c \in \mathbb{R}^{1 \times h}$ => $1 \times h$ param

Nr. param : $g \cdot (dh + hh + h)$ 4 param. din fiecare

$d = 3$

$h = 4$

$n = 2$

$$\Rightarrow n \cdot g(dh + hh + h) = 2 \cdot 4(3 \cdot 4 + 4 \cdot 4 + 4) = 8(12 + 16 + 4) = 2 \cdot 128$$ parameters

total

12. CNN classifier: calculate the number of weights, number of biases, and the size of the associated feature maps.

Assume we have the following CNN classifier:
INPUT, CONV-9-32, POOL-2, CONV-5-64, POOL-2, CONV-5-64, POOL-2, FC-3.
For each layer, calculate the number of weights, number of biases, and the size of the associated feature maps. The size of INPUT is $128 \times 128 \times 3$. The notation follows the convention:

- CONV-K-N denotes a convolutional layer with N filters, each of them of size K×K, padding and stride parameters are always 0 and 1, respectively.
- POOL-K indicates a K×K pooling layer with stride K and padding 0.
- FC-N stands for a fully-connected layer with N neurons.

12. INPUT $128 \times 128 \times 3$

FCN - pag 882
POOL - pag 579
CNN - pag 524

CONV-K-N, POOL-K, FC-N

CONV-9-32 => Kernel size of $9 \times 9 \times 3$, 32 filters
↳ nr. of weights = filter size × nr. filters × nr. channels
$= 9 \times 9 \times 32 \times 3$
$= 7776$
↳ nr. of biases = nr. of filters = 32
↳ output feature map size = (input size - filter size + 2 × padding)/stride + 1
padding = 0
stride = 1
=> output feature map size = input size - filter size + 1
$= 128 - 9 + 1$
$= 120$ => assoc. feat. maps dim: $120 \times 120 \times 3$

POOL - 2 => pooling size of 2×2; no weights or biases, they are 0
↳ output ~~h~~ size = ((input height ~~stride~~ - pooling size + 2 * padding)/stride) + 1
stride = 2
padding = 0
=> output height = $((128 - 2 + 2 \cdot 0)/2) + 1$
$= 64$
output weight = 64
output nr. of channels = input nr. of channels = 3
=> $64 \times 64 \times 3$

FC - 3 => 3 neurons
↳ nr. of weights = input size × nr. of neurons
$= 128 \times 3$
$= 384$
↳ nr. of biases = nr. of neurons = 3

Alin-Beite classifier

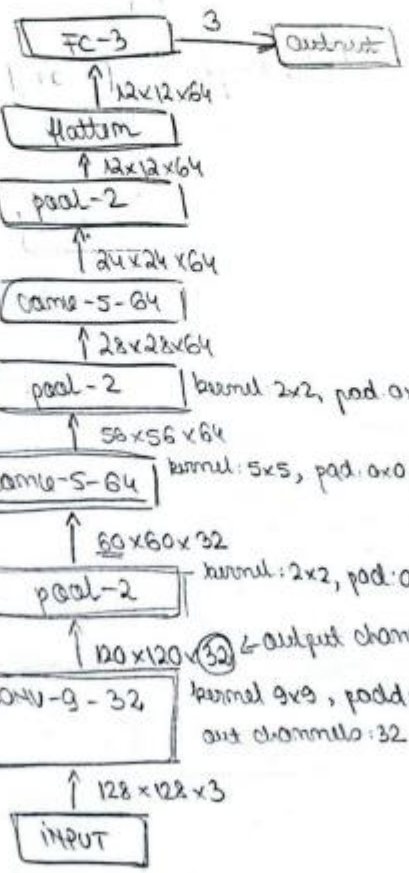
CNN classifier

10) Assume we have the following CNN classifier:

INPUT, CONV-9-32, POOL-2, CONV-5-64, POOL-2, FC-3.

For each layer, calculate the number of weights, number of biases and the size of the associated feature maps. The size of INPUT is 128×128×3. The notation follows the convention:

- CONV-K-N denotes a convolutional layer with N filters, each of size K×K, padding and stride parameters are always 0 and 1
- POOL-K indicates a K×K pooling layer with stride K and padding 0
- FC-N stands for a fully-connected layer with N neurons



W: 12×12×64×3
b: 3

$(24-2+0+2)/2$

$(28-5+0+1)/1$
W: 102400
b: 64 ← output

$(56-2+0+2)/2=28$

$(60-5+0+1)/1=64$
weights: 64·32·5·5 = 51200
bias: 64

$(120-2+0+2)/2=60$
weights: 60·32·2·2=7680
bias: 60

$(32 \times 3) \times 9 \times 9 = 7776$
output input kernel kernel

weights: 7776
bias: 32

! Pt pad min. lenii output latime output

$\lceil (m_h - K_h + P_h + S_h)/S_h \rceil \times \lceil (m_w - K_w + P_w + S_w)/S_w \rceil$

$(128-9+0+1/1=120) \times (128-9+0+1/1=120)$

13. Assume a scalar multiplication and a scalar addition both account for one FLOP and exponentiation accounts for 10 FLOPs.

Assume a scalar multiplication and a scalar addition both account for one FLOP and exponentiation accounts for 10FLOPs. All other scalar operations account for 0 FLOPs. How many FLOPs does the scaled dot-product attention require for n queries and m key-value pairs, where queries and keys length is d and values length is v.

Assume a scalar multiplication and a scalar addition both account for one FLOP and exponentiation accounts for **10** FLOPs. All other scalar operations account for **0** FLOPs. How many FLOPs does the scaled dot-product attention require for n queries and m key-value pairs, where queries and keys are of length d and values are of length v. Give the answer symbolically. Numerical application: $n = 2, m = 3, d = 2, v = 3$.

Solutie:

13. 1 FLOPs - scalar multiplication
scalar addition

10 FLOPs - exponentiation

0 FLOPs - other operation

DOT PROD 757

n - queries

m - key-values

length d

length v

$n = 2$

$m = 3$

$d = 2$

$v = 3$

dot-product attention scoring function: $a(q, k) = q^T k / \sqrt{d}$

$\Rightarrow$ scaled dot-product attention: $\text{Softmax}\left(\frac{Q^T K}{\sqrt{d}}\right) \cdot V \in \mathbb{R}^{n \times v}$

queries

3 values: $Q \in \mathbb{R}^{n \times d}$ (1)

keys $K \in \mathbb{R}^{m \times d}$ (2)

values $V \in \mathbb{R}^{m \times v}$ (3)

$\Rightarrow Q^T \in \mathbb{R}^{d \times n}$ (1)

a P.p. ce avem un query and one key: $a(q, k) = q^T k / \sqrt{d}$

$\Rightarrow q \in \mathbb{R}^{1 \times d}$

$k \in \mathbb{R}^{1 \times d}$ $\Big\} \Rightarrow q^T k \in \mathbb{R}^{d \times d} \Rightarrow d^2$ FLOPs

$q^T k / \sqrt{d}$ $\Big\} \Rightarrow 2d^2$ FLOPs for $a(q, k)$

$\Rightarrow$ for n queries

m key-value pairs $\Big\} \Rightarrow n \cdot m \cdot 2d^2(d-1)$ FLOPs

$= 2 \cdot 3 \cdot 2 \cdot 4(2-1)$

$= 48$ FLOPs

14. Use PCA to reduce the dimension from 2 to 1, to compute the first principal component.

Assume that we have a matrix 4x2 use PCA to reduce the dimension from 2 to 1, to compute the first principal component. Observe that matrix is not centered.

Assume we have the design matrix

$$X = \begin{bmatrix} 6 & -4 \\ -3 & 5 \\ -2 & 6 \\ 7 & -3 \end{bmatrix}.$$

Use PCA to reduce the dimension from **2** to **1**, i.e., to compute the first principal component. Observe that X is not centered.

Solutie:

6) Assume we have the design matrix $X = \begin{bmatrix} 6 & -4 \\ -3 & 5 \\ -2 & 6 \\ 7 & -3 \end{bmatrix}$. Use PCA to reduce the dimension from 2 to 1 to compute the first principal component. Observe that X is not centered.

$\bar{x} = [8 \quad 4] / 4 = [2 \quad 1]$

$$\Sigma = \frac{1}{4}\left[\left(\begin{bmatrix} 6 \\ -4 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right) \cdot \left(\begin{bmatrix} 6 \\ -4 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right)^T + \left(\begin{bmatrix} -3 \\ 5 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right) \cdot \left(\begin{bmatrix} -3 \\ 5 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right)^T + \right.$$

$$\left. + \left(\begin{bmatrix} -2 \\ 6 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right)\left(\begin{bmatrix} -2 \\ 6 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right)^T + \left(\begin{bmatrix} 7 \\ -3 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right)\left(\begin{bmatrix} 7 \\ -3 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right)^T \right] =$$

$$= \frac{1}{4}\left(\begin{bmatrix} 4 & 16 \\ 16 & 64 \end{bmatrix} + \begin{bmatrix} 121 & -11 \\ -11 & 1 \end{bmatrix} + \begin{bmatrix} 100 & -20 \\ -20 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 7 \\ 7 & 49 \end{bmatrix}\right) = \frac{1}{4}\begin{bmatrix} 82 & -20 \\ -20 & 82 \end{bmatrix}$$

pt. verificare: $\tilde{X} = X - \bar{X} = \begin{bmatrix} 4 & -5 \\ -5 & 4 \\ -4 & 5 \\ 5 & -4 \end{bmatrix}$ matricea centrată $= \begin{bmatrix} \frac{41}{2} & -20 \\ -20 & \frac{41}{2} \end{bmatrix}$

$$\Sigma = \frac{1}{m} \cdot \tilde{X}^T \cdot \tilde{X}$$

$\tilde{x} = [0 \quad 0]^T$

14. $X = \begin{pmatrix} 6 & -4 \\ -3 & 5 \\ -2 & 6 \\ 7 & -3 \end{pmatrix}$, X is not centered

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PCA to reduce dim from 2 to 1

Compute first principal component

- columns means: $x_1 = \frac{6-3-2+7}{4} = 2$

$x_2 = \frac{-4+5+6-3}{4} = 1$

$\Rightarrow X_{centered} = \begin{pmatrix} 6 & -4 \\ -3 & 5 \\ -2 & 6 \\ 7 & -3 \end{pmatrix} - \begin{pmatrix} 2 & 1 \\ 2 & 1 \\ 2 & 1 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -5 \\ -5 & 4 \\ -4 & 5 \\ 5 & -4 \end{pmatrix}$

$\Sigma = \frac{1}{n} X^T X = \frac{1}{4} \begin{pmatrix} 4 & -5 & -4 & 5 \\ -5 & 4 & 5 & -4 \end{pmatrix} \begin{pmatrix} 4 & -5 \\ -5 & 4 \\ -4 & 5 \\ 5 & -4 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 82 & -80 \\ -80 & 82 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 41 & -40 \\ -40 & 41 \end{pmatrix}$

$P(\lambda) = \det(\Sigma - \lambda I_2) = \begin{vmatrix} \frac{41}{2} - x & -40 \\ -40 & \frac{41}{2} - \lambda \end{vmatrix} = 0 \Rightarrow \left(\frac{41}{2} - \lambda\right)^2 = (-40)^2$

$\Rightarrow \frac{41}{2} - \lambda = 40 \Rightarrow \lambda = \frac{1}{2}$

$\Rightarrow \frac{41}{2} - \lambda = -40 \Rightarrow \lambda = \frac{81}{2}$ ← Se ia cea mai mare λ

Calc. sist. vezi SUS

$\Sigma w_1 = \lambda w_1 \Rightarrow$ Notăm cu $w_1 = \begin{pmatrix} a \\ b \end{pmatrix}$

$\begin{pmatrix} \frac{41}{2} & -40 \\ -40 & \frac{41}{2} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \frac{81}{2} \begin{pmatrix} a \\ b \end{pmatrix}$

$\begin{cases} \frac{41}{2} a - 20b = \frac{81}{2} a \\ -20a + \frac{41}{2} b = \frac{81}{2} b \end{cases} \Rightarrow \begin{matrix} -20b = 20a \\ b = -a, \ a \in \mathbb{R}^* \end{matrix}$

$\Rightarrow w_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ ← prima componentă principală

15. Values of w and b given by a hard-margin SVM.

Assume we have a training set for two-class classification in one dimension that contains three sample points: point x1 = 3 with label y1 = 1; point x2 = 1 with label y2=1 and point x3=-1 with y3=-1. What are the values of w and b given by a hard-margin SVM?

Assume we have a training set for two-class classification in one dimension that contains three sample points: point $x^{(1)} = 3$ with label $y^{(1)} = 1$, point $x^{(2)} = 1$ with label $y^{(2)} = 1$, and point $x^{(3)} = -1$ with label $y^{(3)} = -1$. What are the values of w and b given by a hard-margin SVM?

Solutie:

pag 128

15. Training set for 2-class classification in one dimension

point: $x^{(1)} = 3$, $x^{(2)} = 1$, $x^{(3)} = -1$

label: $y^{(1)} = 1$, $y^{(2)} = 1$, $y^{(3)} = -1$

hard-margin SVM

$w, b = ?$

- Hard-margin equation: $\min_{w, w_0} \frac{1}{2}\|w\|_2^2$ subject to $y^{(i)}(\underset{w}{w^T} x^{(i)} + \underset{b}{w_0}) \geq 1$, $i = \overline{1,n}$

$$\Rightarrow \begin{cases} 1(w \cdot 3 + b) \geq 1 \\ 1(w \cdot 1 + b) \geq 1 \\ -1(w \cdot (-1) + b) \geq 1 \end{cases} \Leftrightarrow \begin{cases} 3w + b - 1 \geq 0 \\ w + b - 1 \geq 0 \\ +w - b - 1 \geq 0 \end{cases}$$

$$\Rightarrow \begin{cases} 3w + b - 1 \geq 0 \\ w + b - 1 \geq 0 \end{cases} (-)$$

$2w \geq 0 \Rightarrow w \geq 0 \Rightarrow w \in [0, \infty]$

$w - b - 1 \geq 0 \Rightarrow w \in [1, \infty]$ and $b \in [0, \infty]$ și $b < w - 1$

$\Rightarrow$ O soluție posibilă este:

$$\left.\begin{matrix} w = 1 \\ b = 0 \end{matrix}\right\} \Rightarrow \left.\begin{matrix} 3 \cdot 1 + 0 - 1 \geq 0 \\ 1 + 0 - 1 \geq 0 \\ 1 - 0 - 1 \geq 0 \end{matrix}\right\} \Rightarrow \begin{matrix} 2 \geq 0 \ (A) \\ 0 \geq 0 \ (A) \\ 0 \geq 0 \ (A) \end{matrix}$$

?

Rez. 2

15. Training set for 2-class classification in one dim. point:

point: $x^{(1)} = 3$ label: $y^{(1)} = 1$

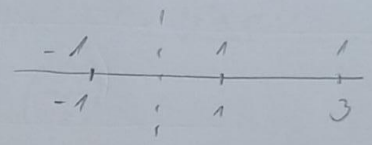
$x^{(2)} = 1$ $y^{(2)} = 1$

$x^{(3)} = -1$ $y^{(3)} = -1$

hard-margin SVM

$w, b = ?$

$$y = \begin{cases} 1 & , wx + b \geq 0 \\ -1 & , wx + b < 0 \end{cases}$$



Se observă că putem alege pentru dr. ce fund vector suport care separă cele 2 clase.

$$\left.\begin{array}{l} y_1 = 1 \Rightarrow x \in \{1, 3\} \Rightarrow \begin{cases} 3w + b \geq 0 \\ w + b \geq 0 \end{cases} \Rightarrow b \geq -w \\ y_1 = -1 \Rightarrow x = -1 \Rightarrow -w + b < 0 \Rightarrow b < w \end{array}\right\} \Rightarrow$$

$\Rightarrow b < w$ și $b \geq -w \Rightarrow w \neq 0 \rightarrow w = 1$

$$\begin{array}{l} b \geq -1 \\ b < 1 \end{array} \Rightarrow b = 0$$

16.Linear regression - batch gradient decent

Assume we have a dataset $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(n)}, y^{(n)})\}$ of points in the 2D plane, and we want to estimate the following linear regression model:

$\hat{y} = w_1 x + w_0$.

Write the batch gradient descent update for w_0, w_1 in terms of $x^{(1)}, y^{(1)}, x^{(2)}, y^{(2)}, \ldots, x^{(n)}, y^{(n)}$, where the learning rate is η. Numerical application: $(1,1), (2,-1), (3,2)$. $\eta = 0.1$.

16. Assume that we have a dataset: $\{(x^{(1)}, y^{(1)}), \ldots (x^{(n)}, y^{(n)})\}$ of points in ~~cont~~ the 2D plane, and we want to estimate the following linear regression model: $\hat{y} = w_1 x + w_0$

Write the batch gradient decent update for w_0, w_1 in terms of $x^{(1)}, y^{(1)}, x^{(2)}, \ldots, x^{(n)}, y^{(n)}$ while the learning rate is η.

Numerical app: $(1,1), (2,-1), (3,2)$, $\eta = 0{,}1$.

$$X = \begin{pmatrix} 1 & x^{(1)} \\ 1 & x^{(2)} \\ 1 & x^{(3)} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad y = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

Gradient descent:

$$\nabla_w \mathcal{L}(w) = \frac{1}{n} \mathcal{L}(w) = \frac{1}{n} X^T (Xw - y)$$

$$w^* = (X^T X)^{-1} X^T y$$

$$\nabla_w \mathcal{L}(w) = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix} \cdot \left[\begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} \cdot \begin{pmatrix} w_0 \\ w_1 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \right]$$

$$= \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} w_0 + w_1 - 1 \\ w_0 + 2w_1 + 1 \\ w_0 + 3w_1 - 2 \end{pmatrix}$$

$$= \frac{1}{3} \begin{pmatrix} 3w_0 + 6w_1 - 2 \\ 6w_0 + 14w_1 - 5 \end{pmatrix}$$

$$w \leftarrow w - \eta \nabla_w \mathcal{L}(w) \qquad \frac{\eta}{n} = \frac{0{,}1}{3}$$

$$\begin{pmatrix} w_0 \\ w_1 \end{pmatrix} \leftarrow \begin{pmatrix} w_0 \\ w_1 \end{pmatrix} - \frac{1}{30} \begin{pmatrix} 3w_0 + 6w_1 - 2 \\ 6w_0 + 14w_1 - 5 \end{pmatrix}$$

$$w_0 \leftarrow w_0 - \frac{3w_0}{30} - \frac{6w_1}{30} + \frac{2}{30}$$

$$w_1 \leftarrow w_1 - \frac{6w_0}{30} + \frac{14w_1}{30} + \frac{5}{30}$$

$$\Rightarrow 30w_0 \leftarrow 30w_0 - 3w_0 - 6w_1 + 2 \Rightarrow w_0 = \frac{-6w_1 + 2}{3}$$

$$30w_1 \leftarrow 30w_1 - 6\,\frac{-6w_1 + 2}{3} - 14w_1 + 5$$

$$12w_1 \leftarrow 5 - 4 \Rightarrow w_1 \leftarrow 0{,}5$$

$$w_0 = \frac{-3+2}{3} = -\frac{1}{3} = -0{,}33$$

$$\Rightarrow \hat{y} = 0{,}5x - 0{,}33$$